

Cycle 1 Clone 1 Report — _infra/pre-registration-gate-policy-scope-verification (Fork 420a6b028dfb)

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Contents

Cycle 1 Clone 1 Report — _infra/pre-registration-gate-policy-scope-verification (Fork 420a6b028dfb)	1
Abstract	1
Verdict	2
Rubric SHA Anchor Chain (Three-Way Byte-Equal)	2
Decision Precedence (Rubric-Verbatim, Off-Disk-Read)	2
Verdict JSON Schema (Well-Formed)	2
Byte-Determinism $\times 2$ (3/3 Artefacts SHA-Equal)	2
Session-Context Matrix	2
Anchor Preservation (18 SHAs; c22 + c46 Unchanged)	3
Discipline Invariants (Grep-Clean)	3
Test Surface	3
Cycle Disposition	3
Ledger Events (12 Shadow Rows Under -clone-1 Suffix)	3
State-Machine Discipline (c29 Lemma Respected)	4
MODERATE Findings (3; All Inherited or Explicitly Disclosed)	4
MINOR Finding (Logged, Not Investigated)	4
Auditor-Discipline Lemma Reinforced (c46 $\rightarrow$ c47)	4
Sub-Topic Assessment	4
Merge Disposition	5
Standing Constraints (Unchanged)	5
Anti-Patterns Locked (5-Count Stable)	5
Cycle-48 Handoff (Seed Carries; Per Cycle-1 Auditor)	5
Cumulative Progress	6

[OUTPUT: report_cycles_1-1_clone_1]

Cycle 1 Clone 1 Report — _infra/pre-registration-gate-policy-scope-verification (Fork 420a6b028dfb)

Abstract

Cycle 1 of clone-1 (fork 420a6b028dfb) closes Branch B _infra/pre-registration-gate-policy-scope-verification at **MIXED**. Empirically verifies the c46 path (ii) amendment's harness-constraint claim: 9 pre-c45 worker-turn commits (all M-SCORE-1/bridge-api-real-audio-quantization/* cycles 38-39) prove path (i) WAS satisfiable under earlier session contexts; 141 harness-boundary commits confirm path (ii) is correct for the current context. The MIXED verdict is mechanically correct against the rubric's decision precedence (auditor re-read rubric off-disk per the c46 auditor-lemma). docs/pre_registration_gate_policy.md §3 append preserves §1+§2 byte-identically (prefix SHA anchor asserts this), documents the partition

without triggering a sunset ticket (rubric text says MIXED does NOT sunset), and leaves future in-turn-capable contexts free to scope-locally reinstate path (i). Auditor decision: **COMPLETE** with `[[BRANCH_COMPLETE]]`.

Verdict

MIXED (VALIDATED; `[[BRANCH_COMPLETE]]` emitted; c46 amendment now formally scope-locked to the harness-boundary bucket for the current session context).

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/pre_registration_gate_policy_scope_verification_rubric.md	1be2bac55ce595b47b6f369f472c3dadff31024d0447c133fd75bdf013251
data/pre_reg_policy_verify/rubric_hash.txt	1be2bac5...11cb
verdict.json.rubric_hash	1be2bac5...11cb

Rubric mtime < scripts mtime (test_01 HARD gate PASS). Rubric committed BEFORE any script under scripts/pre_reg_policy_verify/.

Decision Precedence (Rubric-Verbatim, Off-Disk-Read)

Rubric §“Frozen verdict set” fires **MIXED** iff `worker_in_turn_count > 0 AND periodic_sweep_count > 0`. Observed:

- `worker_in_turn = 9` (all M-SCORE-1/bridge-api-real-audio-quantization/* cycles 38-39)
- `periodic_sweep = 105`
- `unknown = 94` (conservative bucket; see MODERATE #2)

→ **MIXED fires correctly per rubric text**. `verdict.json.decision_rule_applied` matches rubric verbatim.

Verdict JSON Schema (Well-Formed)

Keys = {`verdict`, `rubric_hash`, `counts_by_context`, `evidence_commits_sample`, `decision_rule_applied`} + supplementary `in_turn_total`, `sweep_total`; verdict value MIXED ∈ enum. Test PASS.

Byte-Determinism × 2 (3/3 Artefacts SHA-Equal)

Artefact	SHA-256
<code>commit_classification.tsv</code>	119d41d4... (run_1 == run_2)
<code>session_context_matrix.tsv</code>	81bbf452... (run_1 == run_2)
<code>verdict.json</code>	53febe83... (run_1 == run_2)

Session-Context Matrix

Header + 7 named class rows + TOTAL row; counts sum to 244 (105 + 36 + 9 + 0 + 0 + 0 + 94). Well-formed.

Anchor Preservation (18 SHAs; c22 + c46 Unchanged)

18 SHA entries in data/pre_reg_policy_verify/anchor_preservation.json. 7 c22 stability harness scripts + c46 canonical determinism_check_c46.py recorded with pre==post byte-identity asserted by tests 12 + 13 (both PASS in run auditor executed live).

Policy-doc prefix preservation: first 3488 bytes SHA d432523e... byte-equals pre-edit SHA → §1+§2 preserved verbatim under MIXED-append (new §3 only). This satisfies the “amend, don’t rewrite” contract per c46 lemma.

Discipline Invariants (Grep-Clean)

Zero occurrences of random|secrets\.|numpy\.random|sidecar_nonfactor|i4_stratified under scripts/pre_reg_policy_verify/. Interpreter guard /usr/bin/python3 on every new script.

Test Surface

Suite	Result
tests/test_pre_reg_policy_verify.py	15/15 PASS (exceeds ≥12 minimum)
tests/test_integration_cross_branch.py §62 (checks a-h)	8/8 PASS (lines 4599-4688)
python3 -m long_exposure.tools.promise_check .	NOT MET — pre-existing c46 ERROR at ledger line 745 (not Branch B’s; see MODERATE #1)

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric	Rubric + scripts + grep git-log commit classification + session-context matrix + verdict.json + tests + report + §3 policy-doc append (MIXED clause) + 12 shadow-ledger events	COMPLETE with [[BRANCH_COMPLETE]]

Ledger Events (12 Shadow Rows Under -clone-1 Suffix)

6 named + 2 housekeeping + M-INGEST-1 egress + additional infra events. All 12 events landed in shadow ledger at /home/user/music-gen-instance/fork-420a6b028dfb/clone-1/promise_ledger.jsonl (resolved via long_exposure.workspace_bootstrap.resolve_ledger_path — the correct fanout location for a clone; concat to main happens at post-merge). Sequence matches brief §10 order under -clone-1 suffix.

Key events include: - _infra/pre-registration-gate-policy-scope-verification-clone-1 (in-progress + validated-verdict-rollup) - _plan/rubric-committed-clone-1 - Sub-leaf events for grep, classification, verdict-emission - M-INGEST-1/egress-probe-cycle47-clone-1 (429 + tv_embedded

failure — non-blocking; not the two-consecutive-media_ok=true unblock signal) - _run/cycle_47_
closed-clone-1, _archive/cycle-47-scratch-clone-1

State-Machine Discipline (c29 Lemma Respected)

_infra/pre-registration-gate-policy-scope-verification is a peer sub-milestone under root infra chain (c14/c22/c32/c33 hardening ancestry). NOT a child of any terminal-validated ancestor. Extends the c14 _ledger_schema.py → c22 v2 schema → c33 guard + c36 v2 writer → c39 v3 doc → c46 amendment chain by empirically verifying the c46 amendment's scope.

MODERATE Findings (3; All Inherited or Explicitly Disclosed)

1. **Inherited promise_check ERROR at ledger line 745** (_manager/M-EAR-1-real-label-training-v2-unfixable-by-audit-clone-1 uses supersedes_path where writer expects supersedes). Confirmed off-disk; c46 event (ts 2026-08-29T17:20:01Z), NOT Branch B's. Worker correctly flagged as c48 handoff #6. Merge cycle should either (a) amend c46 row to use supersedes or (b) relax promise_check since supersedes_path is documented as valid variant in _infra/ledger-schema-hardening-v2.
2. **94 unknown-bucket commits** (39% of the classified total) are Add music-gen run artifacts ... variants missing the (periodic sweep) / (merge ...) envelope. Report §9 discloses conservatively; reclassification would only strengthen MIXED toward CONFIRMED-for-current-context, never overturn (the 9 worker-turn c38/c39 commits are independent). Acceptable disclosure; merge cycle may want one regex rule to _MARKER_RULES in classify_commits.py to raise classification precision — non-blocking.
3. **Merge-report path fallback:** /home/user/music-gen-instance/fork-420a6b028dfb/clone-1/merge_report.md target blocked by sandbox (empirically re-confirmed). Worker wrote to workspace root merge_report_c47_branch_b_clone_1.md, matching c39/c40/c41 precedent. Root conductor knows to pick up from workspace root. Non-blocking but worth naming a durable _infra/merge-report-sandbox-fallback-convention at campaign level.

MINOR Finding (Logged, Not Investigated)

- **87 pre-existing cross-branch integration failures** in unrelated sections (environmental drift documented by c46 restored context). Not touched by Branch B. Out of scope for this audit.

Auditor-Discipline Lemma Reinforced (c46 → c47)

Reading rubric docs off-disk (not from brief paraphrases) prevented over-flagging this cycle. The rubric's MIXED clause matches verdict.json.decision_rule_applied string exactly. This is the c46 auditor-lemma applied here; reinforces as durable auditor discipline going forward.

Sub-Topic Assessment

Brief Exit Gate	Status
Rubric mtime < scripts (test 01 HARD)	MET
verdict ∈ {CONFIRMED, LIFTED, MIXED} + three-way rubric_hash byte-equality	MET (MIXED, hash 1be2bac5...)
Byte-determinism × 2 SHA-equal on classification + matrix + verdict	MET

Brief Exit Gate	Status
Anchor preservation: c22 + c46 unchanged ≥12 test cases green	MET (tests 12 + 13 PASS) MET (15/15)
Cross-branch §62 extended with ≥8 checks 0-ERROR promise_check	MET (8/8) NOT MET (pre-existing c46 ERROR at line 745; not Branch B's — MODERATE #1)
6 named + 2 housekeeping + M-INGEST-1 egress ledger events under -clone-1 on infra families	MET (12 total in shadow ledger)
Report with 10-section template + §10 c48 handoff seeds	MET
Merge report at fanout path	PARTIALLY MET (sandbox blocks target; workspace-root fallback per c41 precedent)

Merge Disposition

Merge report at workspace root merge_report_c47_branch_b_clone_1.md per documented c39/c40/c41 workaround (sandbox denies writes to /home/user/music-gen-instance/...).

Merge-cycle tasks: 1. Concat clone-1 shadow ledger (12 events) into main promise_ledger.jsonl. 2. Register _infra/pre-registration-gate-policy-scope-verification-clone-1 + 5 named sub-leaves in plan_of_record.md Milestones table (pattern per c38/c44/c46 post-merge integrations). 3. Address MODERATE #1: amend c46 event to use supersedes OR relax promise_check per _infra/ledger-schema-hardening-v2. 4. (Optional, non-blocking) One regex rule to _MARKER_RULES in classify_commits.py to reduce unknown-bucket size.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908 (not relevant to this branch).
- SHA-256 tiebreak; no PRNG; no sidcar_nonfactor; no i4_stratified (grep-verified).
- Interpreter guard #!/usr/bin/python3 on every new script.
- Read-only anchors preserved: c14 _ledger_schema.py; c22 stability harness (7 scripts); c32/c33/c36-v2/c39-v3 fanout-namespace chain; c46 canonical determinism_check_c46.py; docs/pre_registration_gate_policy.md §1+§2 (prefix SHA d432523e... byte-preserved).
- Rated audio egress-blocked at *.googlevideo.com (429 + tv_embedded closure; retry cadence at conductor level; not required — analytical infra codification).
- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses
UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).

Anti-Patterns Locked (5-Count Stable)

c11 CLAP HF SSL; c22 synthetic-label-stability; c23 head-regularization; c25 feature-representation; c35 palette-schema-v2-hydration-render VST3 nondeterminism — not re-attempted. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged. c31 STILL_GAP surface intact.

No new anti-patterns from this branch; 5-entry campaign registry unchanged.

Cycle-48 Handoff (Seed Carries; Per Cycle-1 Auditor)

1. **Rubric-doc-not-brief-paraphrase auditor discipline lemma** (from c46 audit, further validated this cycle) — codify as durable auditor guidance.

2. **Fold the c46 supersedes_path ERROR at line 745** — either amend the event or expand `promise_check` acceptance.
3. **_infra/merge-report-sandbox-fallback-convention** codification for durable future clones (fourth+ observation of the sandbox-write refusal pattern; c39/c40/c41 + this cycle).
4. **Merge integration for Branch B** — concat + plan-of-record registration.
5. **Fanout-cadence continuation** — Branches A (v2.1) and C (deprecation/anchor pin) are separate clone contexts audited independently.

Cumulative Progress

Infra-hardening chain (post-c47 Branch B close):

Cycle	Milestone	Status
c14	SSoT_ledger_schema.py	landed
c22	v2 schema extension	landed
c33 guard + c36 v2 writer	fanout-namespace codification	field-tested through c37 + c38 (two 3-clone fanouts, zero LedgerConcatError)
c39 v3 doc	codifies auto-suffix-all behaviour	CONVENTION_v3_LANDS
c46	pre-registration gate policy amendment (path (ii))	landed
c47 Branch B (this)	empirical scope-verification of c46 amendment	MIXED (harness-constraint claim scope-locked to current session context; path (i) preserved as documented option for future in-turn-capable contexts)

Pre-registration gate policy arc: c46 amendment (path (ii)) → c47 Branch B empirical MIXED verdict → policy doc §3 appends the scope partition. Amendment claim now scope-locked to the harness-boundary bucket for the current session context. The c46 amendment survives as correct-for-context; path (i) remains as a documented option for future contexts falling into the in-turn-capable bucket.

Ledger state (main): 752 rows, latest c46. Clone-1 shadow ledger: 12 rows staged for post-merge concat. Expected post-merge total: ~764.

Auditor discipline lemma reinforced (c46 → c47): reading rubric docs off-disk (not from brief paraphrases) prevented over-flagging this cycle. Durable auditor guidance going forward.

Egress: still 429 + `tv_embedded` failure mode; c47 Branch B M-INGEST-1/egress-probe-cycle47-clone-1 row records the current failure honestly. Not the two-consecutive-media_ok=true unblock signal.

Fanout cadence: c47 delivered 3-branch fanout (A/B/C). This audit closes Branch B. Branches A (v2.1) and C (deprecation/anchor pin) are separate clone contexts audited independently.

c29 state-machine lemma respected: peer sub-milestone; ledger topology stays a DAG.

c32 → c33 → c36 v2 → c39 v3 fanout-namespace convention held: 12 events correctly emitted under `-clone-1` suffix.

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Scope of this fanout clone (Branch B / _infra/pre-registration-gate-policy-scope-verification) is fully discharged. [[BRANCH_COMPLETE]] emitted per auditor role definition on validated + scope-exhausted branch.

[END OUTPUT]